

Compiled with problems:

ERROR in ./docs/semester-1/gqur-300/unit-03/topic-01.mdx

Module build failed (from ./node_modules/@docusaurus/mdx-loader/lib/index.js):

Error: MDX compilation failed for file

["/home/a2ahs/mega_book_for_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx"](/home/a2ahs/mega_book_for_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx)

Cause: Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a letter, `\$`, or `\_`

Details:

```
{
  "column": 36,
  "file": "",
  "message": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "line": 141,
  "name": "141:36",
  "place": {
    "_bufferIndex": 33,
    "_index": 0,
    "line": 141,
    "column": 36,
    "offset": 7642
  },
  "reason": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "ruleId": "unexpected-character",
  "source": "micromark-extension-mdx-jsx",
  "url": "https://github.com/micromark/micromark-extension-
mdx-jsx#unexpected-character-at-expected-expect"
}
```

at compileToJSX

4. 12×0.1 equals: (*Apply*) a) 120 b) 12 c) 1.2 d) 0.12
5. The ratio 40 girls to 20 boys in its simplest form is: (*Understand*) a) 40 to 20 b) 2 to 1 c) 1 to 2 d) 20 to 40
6. A recipe for 12 people uses 4 eggs. For 30 people the recipe needs: (*Apply*) a) 8 eggs b) 10 eggs c) 12 eggs d) 16 eggs
7. After a 10 percent rise, a school's electricity bill is 17,600 rupees. The bill before the rise was: (*Apply*) a) 15,840 rupees b) 16,000 rupees c) 17,390 rupees d) 19,360 rupees
8. 8 squared equals: (*Remember*) a) 16 b) 64 c) 82 d) 512
9. The square root of 81 is: (*Apply*) a) 8 b) 9 c) 40.5 d) 41
10. 187 pupils must travel in vans of 26 seats. Since 187 divided by 26 is 7 with remainder 5, the school needs: (*Apply*) a) 7 vans b) 8 vans c) 5 vans d) 26 vans

Restricted-response questions (RRQs)

1. Order these from smallest to largest: 2, -3, $\frac{1}{2}$, 0.75, -0.5. (*Apply*)
2. A shop's cash box holds 9,000 rupees and its debts total 4,500 rupees. Write the shop's position using integers and compute it. (*Apply*)
3. Explain in two sentences why 12×0.1 is smaller than 12, using the idea of a part of a whole. (*Understand*)
4. A recipe for 20 people uses 6 spoons of sugar. How much sugar for 50 people? Name the multiplier you used. (*Apply*)
5. Compute 25 percent off a 640-rupee atlas in two ways (subtract the discount; pay 75 percent directly) and confirm the two answers agree. (*Apply*)
6. A friend scales a recipe by adding: "20 more people, so add 6 more spoons of sugar." Explain the error in two sentences and state what to do instead. (*Analyze*)
7. Between which two whole numbers does the square root of 50 lie? Justify your answer in one sentence. (*Apply*)

8. For each situation, name the operation and give your reason: (a) 15 boxes of 40 books each; (b) 300 pupils placed into classes of 30. (*Apply*)
9. Compute 7 squared and the square root of 100, then explain in one sentence how the tile picture connects squaring and square roots. (*Understand*)
10. A trainee computes 187 divided by 26 as 7 and orders 7 vans. Identify the error in one sentence using the look-back idea. (*Analyze*)

Extended-response questions (ERQs)

1. A school canteen buys flour in 5.5 kg bags at 1,200 rupees per bag. One batch of parathas uses 0.75 kg. Compute how many full batches one bag gives and the flour left over, compute the flour cost per batch to the nearest rupee, and check both answers for reasonableness using estimation, showing your check. (*Analyze*)
2. A stationery supplier offers two deals on notebooks costing 80 rupees each: Deal A gives "20 percent off", Deal B gives "5 notebooks free with every 20 bought". The school needs 100 notebooks. Compute the total cost under each deal, state which is cheaper and by how much, and identify one circumstance under which the school might still prefer the other deal, with a reason. (*Analyze*)
3. The school wants a square mosaic of 225 small tiles on a corridor wall, and the wall space is a square of side 20 tiles. State how wide the mosaic is and whether it fits; the school has only these 225 tiles in stock, so compute how many more tiles a full 20-by-20 design would need; and propose one real reason a school might prefer the smaller mosaic, with a computation that supports it. (*Analyze*)
4. Five pupils scored 36/45, 42/60, 55/75, 18/25 and 27/30 on different tests. Convert each to a percentage, rank the pupils, and analyse which raw score looked most impressive but performed worst, explaining why the total matters when comparing results. (*Analyze*)
5. A school must transport 187 pupils to an event. Each van carries 26 passengers and costs 3,000 rupees; the transport budget is 25,000 rupees. Compute the vans needed (with your look-back check), the total cost, and whether the budget fits. Then evaluate one alternative (for example, a larger 40-seat bus at 4,500 rupees) and recommend a plan with reasons. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. **a** - whole numbers are 0 and the counting numbers; the other lists contain a negative, a fraction, or a decimal.
2. **b** - the negative sign records that the amount reduces the shop's position; that is bookkeeping made honest.
3. **a** - 5.5 divided by 2.75 is exactly 2.
4. **c** - multiplying by 0.1 takes one tenth of 12: 1.2.
5. **b** - 40 to 20 simplifies by dividing both by 20: 2 to 1.
6. **b** - the multiplier is $30/12 = 2.5$, and $4 \times 2.5 = 10$.
7. **b** - the rise acted on the original, so divide: $17,600 / 1.1 = 16,000$.
8. **b** - $8 \times 8 = 64$.
9. **b** - $9 \times 9 = 81$.
10. **b** - 7 vans carry 182 pupils and 5 remain; the real-world answer rounds up.

RRQ model answers and mark schemes

1. Model: -3, -0.5, $1/2$, 0.75, 2. **Marks (2):** 1 - negatives placed before positives; 1 - full correct order.
2. Model: position = $9,000 + (-4,500) = 4,500$ rupees. **Marks (2):** 1 - correct integer expression; 1 - correct value.
3. Model: 0.1 is one tenth, so the product takes one tenth of 12, and taking a part gives less than the whole. **Marks (2):** 1 - names 0.1 as a part (tenth); 1 - connects part to shrinking.
4. Model: multiplier $50/20 = 2.5$; sugar = $6 \times 2.5 = 15$ spoons. **Marks (2):** 1 - correct amount; 1 - multiplier named.
5. Model: discount = $0.25 \times 640 = 160$; price = $640 - 160 = 480$. Direct: $0.75 \times 640 = 480$. Both give 480 rupees. **Marks (2):** 1 - both routes computed; 1 - agreement confirmed.
6. Model: the taste depends on the ratio of ingredients, and adding keeps the difference constant while destroying the ratio; the fix is to multiply every ingredient by the crowd multiplier (here 2 for 40 people). **Marks (2):** 1 - the ratio-vs-difference point; 1 - the multiply fix.

7. Model: between 7 and 8, because $7 \times 7 = 49 < 50 < 64 = 8 \times 8$. **Marks (2):** 1 - correct pair; 1 - justification via neighbouring squares.
8. Model: (a) multiply - the total combines 15 equal groups of 40; (b) divide - the total breaks into equal groups of 30. **Marks (4):** 1 - (a) multiply; 1 - reason; 1 - (b) divide; 1 - reason.
9. Model: $7 \text{ squared} = 49$; $\text{sqrt}(100) = 10$. The tile picture: a square floor of side 7 holds 49 tiles (squaring climbs from side to area), and the root climbs back from area to side. **Marks (3):** 1 - both values; 2 - the connection explained (1 for direction, 1 for the side-area idea).
10. Model: the look-back check shows 7 vans carry only 182 of 187 pupils, so the discarded remainder of 5 pupils forces an 8th van. **Marks (2):** 1 - identifies the discarded remainder; 1 - states the corrected answer.

ERQ rubrics

1. Canteen flour. Analytic rubric: *Division correct* (0-2: 7 batches, 0.25 kg left); *Cost per batch* (0-2: $1,200 / 7$ gives about 171 rupees per batch, or an equivalent defensible computation shown); *Reasonableness check* (0-2, **Analyze**: an explicit estimation check, for example that 5.5 kg is a little over $7 \times 0.75 = 5.25$, so 7 batches is plausible). Total 6.
2. Notebook deals. Analytic rubric: *Both totals* (0-2: Deal A = $100 \times 80 \times 0.8 = 6,400$ rupees; Deal B = pay for 80, receive 100 = 6,400 rupees); *Comparison* (0-2: the deals tie, and noticing the tie earns full marks); *Circumstance analysed* (0-2, **Analyze**: a real non-price factor such as invoice timing, storage, or supplier reliability argued with a reason). Total 6.
3. Mosaic. Analytic rubric: *Root and fit* (0-2: 15 by 15 fits within 20 by 20); *Tile count* (0-2: 400 tiles needed, 175 more required); *Supported proposal* (0-2, **Analyze**: a reason argued with a computation, for example spare tiles for future repairs or the cost of 175 extra tiles). Total 6.
4. Marksheet comparison. Analytic rubric: *Conversions* (0-3: 80, 70, 73.3, 72, 90 percent, 1 mark per two correct); *Ranking* (0-1: 27/30 first down to 42/60 last); *Analysis* (0-3, **Analyze**: identifies 42/60 as the impressive-looking but weakest score and explains that the total renames the share per hundred, so raw sizes cannot be compared across totals). Total 7.
5. Transport plan. Analytic rubric: *Vans with look-back* (0-2: 8 vans, with the remainder check shown); *Cost and budget* (0-2: $8 \times 3,000 = 24,000$ rupees, fits within 25,000); *Alternative evaluated* (0-3, **Analyze**: a real alternative

computed, for example 5 vans of 40 = 200 seats, $5 \times 4,500 = 22,500$ rupees, cheaper with fewer vehicles); *Recommendation* (0-2: a justified choice between the plans); *Clarity* (0-1). Total 10.

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

Give feedback